MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC TEST – 05 (POLYNOMIALS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM:** 30

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

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Q1. If p and q are the zeroes of the polynomial $x^2 + px + q$ where $q \neq 0$, then: [1]

(A) $p = 1,\ q = -2$ **(B)** $p = -1,\ q = 2$

(C) $p = 1,\ q = 2$ **(D)** $p = -2,\ q = 1$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

We know Vieta's relations for zeroes p and q of $x^2 + px + q$:

Formula Used: Sum $= p + q = -p$, Product $= pq = q$

Since $q \neq 0$, dividing $pq = q$ by q yields $p = 1$. Substituting $p = 1$ into $p + q = -p$: $1 + q = -1 \implies q = -2$. Hence $\boxed{p = 1,\ q = -2}$. [1M]

Q2. The product of the zeroes of the polynomial $mx^2 + 6x + (2m - 1) = 0$ is -1. Then $m =$: [1]

(A) $\frac{1}{3}$ (B) $-\frac{1}{3}$ (C) 1 (D) -1

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Comparing $mx^2 + 6x + (2m - 1)$ $(m \neq 0)$ with $ax^2 + bx + c$, we have $a = m$ and $c = 2m - 1$. We know:

Formula Used: Product of zeroes $= \frac{c}{a} = -1$

Thus $\frac{2m-1}{m} = -1 \implies 2m - 1 = -m \implies 3m = 1 \implies \boxed{m = \frac{1}{3}}$. [1M]

Q3. If $\alpha+\beta=3$ and $\alpha^3+\beta^3=7$, then α,β are the zeroes of the quadratic polynomial: [1]

(A) $9x^2-27x+20$ (B) $3x^2+9x-7$ (C) $2x^2-6x+15$ (D) $9x^2+27x-20$

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

We know the cubic identity $\alpha^3+\beta^3=(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)$:

Formula Used: $p(x)=k\left[x^2-(\alpha+\beta)x+\alpha\beta\right]$

Substituting: $7=3^3-3\alpha\beta(3) \implies 7=27-9\alpha\beta \implies 9\alpha\beta=20 \implies \alpha\beta=\frac{20}{9}$. Choosing $k=9$ gives $9\left(x^2-3x+\frac{20}{9}\right)=\boxed{9x^2-27x+20}$. [1M]

Q4. If α,β are zeroes of x^2-px+q, then the quadratic polynomial with $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ as zeroes is: [1]

(A) qx^2-px+1 (B) qx^2+px+1 (C) px^2-qx+1 (D) x^2-qx+p

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

From x^2-px+q, $\alpha+\beta=p$ and $\alpha\beta=q$. We know:

Formula Used: $S'=\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}, \qquad P'=\frac{1}{\alpha\beta}$

Thus $S'=\frac{p}{q}$ and $P'=\frac{1}{q}$, giving $k\left(x^2-\frac{p}{q}x+\frac{1}{q}\right)$; taking $k=q$ yields $\boxed{qx^2-px+1}$. [1M]

Q5. Assertion-Reason: [1]

Assertion (A): If one zero of $4x^2-8kx-9$ is negative of the other, then $k=0$.

Reason (R): For negative zeroes, $\alpha+(-\alpha)=0 \implies -\frac{b}{a}=0 \implies b=0$.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) Both A and R are true and R is the correct explanation of A.

We know Vieta's sum formula when zeroes are opposites (α and $-\alpha$):

Formula Used: $\alpha+(-\alpha)=-\frac{b}{a}=0 \implies b=0$

Here $a=4$, $b=-8k$, so $-8k=0 \implies k=0$. Both **(A)** and **(R)** (interpreting "negative zeroes" as zeroes negative of each other) are true and **(R)** explains **(A)**. [1M]

Q6. If one zero of $(a^2+9)x^2+13x+6a$ is reciprocal of the other, find the value of a. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let the zeroes be α and $\frac{1}{\alpha}$. We know Vieta's product relation:

Formula Used: $\alpha\beta = \dfrac{C}{A}$

Here $A = a^2+9$ and $C = 6a$, so $\frac{6a}{a^2+9} = 1 \implies a^2-6a+9=0 \implies (a-3)^2=0 \implies \boxed{a=3}$. [2M]

Q7. If α, β are the zeroes of $p(x) = x^2+x+1$, find the value of $\dfrac{1}{\alpha}+\dfrac{1}{\beta}$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

From $p(x) = x^2+x+1$, we have $\alpha+\beta=-1$ and $\alpha\beta=1$. We know:

Formula Used: $\dfrac{1}{\alpha}+\dfrac{1}{\beta} = \dfrac{\alpha+\beta}{\alpha\beta}$

Substituting the values:

$$\frac{1}{\alpha}+\frac{1}{\beta} = \frac{-1}{1} = \boxed{-1}. \qquad [2M]$$

Q8. If α and β are the zeroes of $f(x) = 2x^2+5x+k$ such that $(\alpha+\beta)^2-\alpha\beta = \dfrac{21}{4}$, determine k. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

From $f(x) = 2x^2+5x+k$, we have $a=2, b=5, c=k$. We know Vieta's relations:

Formula Used: $\alpha+\beta = -\dfrac{5}{2}, \qquad \alpha\beta = \dfrac{k}{2}$

Substituting into $(\alpha+\beta)^2-\alpha\beta = \frac{21}{4}$:

$$\left(-\frac{5}{2}\right)^2 - \frac{k}{2} = \frac{21}{4} \implies \frac{25}{4} - \frac{k}{2} = \frac{21}{4} \implies \frac{k}{2} = \frac{4}{4} = 1 \implies \boxed{k=2}. \qquad [2M]$$

Q9. Find a quadratic polynomial whose zeroes are $2+\sqrt{5}$ and $2-\sqrt{5}$, and verify that the sum and product match the coefficients. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let $\alpha = 2+\sqrt{5}$ and $\beta = 2-\sqrt{5}$. We know:

Formula Used: $p(x) = k(x^2 - Sx + P), \qquad \alpha+\beta = -\frac{b}{a}, \qquad \alpha\beta = \frac{c}{a}$

Step 1 – Construction:

$$S = (2+\sqrt{5}) + (2-\sqrt{5}) = 4, \qquad P = (2+\sqrt{5})(2-\sqrt{5}) = 4-5 = -1. \qquad \textbf{[1M]}$$

Taking $k=1$, the required polynomial is $\boxed{p(x) = x^2 - 4x - 1}$. **[1M]**

Step 2 – Coefficient Verification: Here $a=1, b=-4, c=-1$:

$$-\frac{b}{a} = -\frac{(-4)}{1} = 4 = S, \qquad \frac{c}{a} = \frac{-1}{1} = -1 = P. \qquad \textbf{[1M]}$$

Q10. If α, β are zeroes of $x^2 - 5x + k$ such that $\alpha - \beta = 1$, find the value of k. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

From $x^2 - 5x + k$, we know Vieta's relations and the identity:

Formula Used: $\alpha+\beta = 5, \quad \alpha\beta = k, \qquad (\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$

Solving $\alpha+\beta = 5$ and $\alpha-\beta = 1$ simultaneously: adding gives $2\alpha = 6 \implies \alpha = 3$, and substituting back gives $\beta = 5-3 = 2$. **[2M]**

Therefore $k = \alpha\beta = (3)(2) = \boxed{6}$ (equivalently $5^2 - 4k = 1^2 \implies 4k = 24 \implies k = 6$). **[1M]**

Q11. If the sum of the zeroes of $f(x) = 2x^2 - 3kx + 5$ is 6, find the value of k and the product of zeroes. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Comparing $f(x) = 2x^2 - 3kx + 5$ with $ax^2 + bx + c$, we have $a=2, b=-3k, c=5$. We know:

Formula Used: Sum of zeroes $= -\frac{b}{a}$, Product of zeroes $= \frac{c}{a}$

Using sum of zeroes $= 6$:

$$-\frac{(-3k)}{2} = 6 \implies \frac{3k}{2} = 6 \implies 3k = 12 \implies \boxed{k = 4}. \qquad \textbf{[2M]}$$

And the product of zeroes is:

$$\alpha\beta = \frac{c}{a} = \boxed{\frac{5}{2}}. \qquad \textbf{[1M]}$$

Q12. If the roots of the equation $(x-b)(x-c)+(x-c)(x-a)+(x-a)(x-b)=0$ are equal real numbers, prove that $a=b=c$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Expanding each quadratic term in $(x-b)(x-c)+(x-c)(x-a)+(x-a)(x-b)=0$:

$$[x^2-(b+c)x+bc]+[x^2-(c+a)x+ca]+[x^2-(a+b)x+ab]=0$$
$$\implies 3x^2-2(a+b+c)x+(ab+bc+ca)=0. \quad \textbf{[1.5M]}$$

Comparing with $Ax^2+Bx+C=0$, we have $A=3$, $B=-2(a+b+c)$, $C=ab+bc+ca$. We know:

Formula Used: Equal real roots $\iff D=B^2-4AC=0$

Setting $D=0$:

$$\begin{aligned}[-2(a+b+c)]^2-4(3)(ab+bc+ca)&=0\\ 4(a^2+b^2+c^2+2ab+2bc+2ca)-12(ab+bc+ca)&=0\\ 4(a^2+b^2+c^2-ab-bc-ca)&=0\\ 2(2a^2+2b^2+2c^2-2ab-2bc-2ca)&=0 \quad \textbf{[2M]}\\ 2\big[(a-b)^2+(b-c)^2+(c-a)^2\big]&=0.\end{aligned}$$

Because a,b,c are real numbers, each squared term is ≥ 0. A sum of non-negative terms vanishes only if each term vanishes individually:

$$a-b=0,\quad b-c=0,\quad c-a=0 \implies \boxed{a=b=c.} \text{ (Hence Proved)} \quad \textbf{[1.5M]}$$

Q13. If the zeroes of the polynomial x^2-bx+c are two consecutive positive integers, prove that $b^2-4c=1$. Hence find the zeroes if $b=5$. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let the consecutive positive integer zeroes be $\alpha=n$ and $\beta=n+1$ ($n\in\mathbb{Z}^+$), so $\beta-\alpha=1$. Comparing x^2-bx+c with Ax^2+Bx+C:

$$\alpha+\beta=-\frac{(-b)}{1}=b, \qquad \alpha\beta=\frac{c}{1}=c. \quad \textbf{[1M]}$$

We know the square-difference identity:

Formula Used: $(\beta-\alpha)^2=(\alpha+\beta)^2-4\alpha\beta$

Part 1 – Proof: Substituting $\beta-\alpha=1$, $\alpha+\beta=b$, and $\alpha\beta=c$:

$$1^2=b^2-4c \implies \boxed{b^2-4c=1.} \text{ (Hence Proved)} \quad \textbf{[2M]}$$

Part 2 – Zeroes when $b=5$: Substituting $b=5$ into $b^2-4c=1$:

$$5^2-4c=1 \implies 25-4c=1 \implies 4c=24 \implies c=6. \quad \textbf{[1M]}$$

Thus the polynomial becomes $x^2-5x+6=(x-2)(x-3)=0 \implies$ Zeroes: $\boxed{2 \text{ and } 3}$. **[1M]**

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